

Research interests

Machine learning in finance, large language models, random matrix theory, kernel methods, financial econometrics

Education

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland
September 2023 – present

PhD in Machine Learning applied to Finance

Advisor: Prof. Semyon Malamud (Swiss Finance Institute, EPFL)

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland
September 2019 – August 2021

Master of Science in Financial Engineering, GPA: 5.8/6

Master's thesis: Use of machine learning techniques for portfolio insurance (at Vaudoise Assurances)

University of Lausanne (UNIL), Faculty of Business and Economics (HEC), Lausanne, Switzerland
September 2016 – June 2019

Bachelor of Science (BSc) in Economics, GPA: 5.6/6

Publications

Scaling Point-in-Time Language Models

NeurIPS 2026

with Bryan Kelly, Semyon Malamud and Teng Andrea Xu

arXiv:2607.11889 • NBER WP 35247

- Shows that training decoder-only transformers up to 4B parameters on 1T chronologically filtered tokens largely closes the performance gap between lookahead-free language models and their unconstrained counterparts

To be presented at: NeurIPS 2026 (poster), EPFL Computational Future seminar (F. Kaplan)

Working papers

Training NTK to Generalize with KARE

2025

with Bryan Kelly, Semyon Malamud and Teng Andrea Xu

arXiv:2505.11347 • SFI Research Paper 25-51

- Proposes optimizing the neural tangent kernel explicitly rather than implicitly through gradient descent, yielding kernels that match or outperform the deep networks they are derived from

A Test of the Efficiency of a Given Portfolio in High Dimensions

2025

with Mikhail Chernov, Bryan Kelly and Semyon Malamud

NBER WP 33565 • SFI Research Paper 25-26

- Extends the Gibbons–Ross–Shanken test to settings where test assets far outnumber the sample, using Random Matrix Theory to correct the bias induced by regularized covariance estimation

Presented at: SoFiE 2025 (Paris), ESSFM 2025 (Gerzensee)

Teaching

Teaching Assistant, EPFL, Master in Financial Engineering

September 2023 – present

- Stochastic Calculus
- Machine Learning in Finance

Honors and awards

- **Best Teaching Assistant, 2026** — awarded by student vote in the EPFL Master in Financial Engineering, for Stochastic Calculus and Machine Learning in Finance
- **Swiss Life Award, 2021** — awarded to the student graduating with the highest GPA in the EPFL Master of Science in Financial Engineering

Industry experience

Quantitative Analyst, Taranis – Hedge Fund, Geneva, Switzerland

March 2022 – May 2023

- Designed a backtesting environment for option-based strategies on a SQL database of over 1 billion data points, developed a Python library for the statistical analysis of trading signals, and co-designed an automated trading bot

Asset Management Intern, Vaudoise Assurances, Lausanne, Switzerland

March 2021 – December 2021

- Calibrated an option-based portfolio insurance (OBPI) strategy and built Python tools for option pricing, backtesting and reporting

Skills

- | | |
|-------------------------------------|---|
| • Programming | Python (PyTorch, NumPy, pandas, scikit-learn), MATLAB, SQL, $\text{\LaTeX}$ |
| • High-performance computing | Multi-node GPU training (128 GPUs at CSCS), MPI, NCCL, SLURM |

Languages

- | | |
|------------------|-------------------------|
| • French | Native |
| • English | Fluent (C1) |
| • German | Upper intermediate (B2) |

Last updated: October 8, 2026